

Mathematical proof - Set 1

TMUA practice questions by topic. Specification Prf1-Prf4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Consider the claim: **for every positive integer n , $n^2 + n$ is even**. Which of the following is a complete and valid proof?
- A** $n^2 + n$ is even because both n^2 and n are even.
- B** If n is even then n^2 is even and so $n^2 + n$ is even.
- C** Suppose $n^2 + n$ is odd. Then n is odd, so n^2 is odd, so $n^2 + n$ is even, which is a contradiction... unless n is even.
- D** $1^2 + 1 = 2$, $2^2 + 2 = 6$, $3^2 + 3 = 12$, which are all even, so the claim holds.
- E** $n^2 + n = n(n + 1)$ is the product of two consecutive integers, one of which must be even, so the product is even.

- 2 Which of the following is a counterexample to the claim '**if n is a positive integer, then $2^n + 1$ is prime**'?

- A** $n = 8$
- B** $n = 1$
- C** $n = 4$
- D** $n = 3$
- E** $n = 2$

- 3 Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.

II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.

III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.

IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

- A** II, I, IV, III
- B** I, II, IV, III
- C** II, IV, I, III
- D** I, IV, II, III
- E** II, I, III, IV

- 4 A student wants to prove that **the square of any odd integer leaves remainder 1 when divided by 8**. Which of the following is the most appropriate first step?
- A Check the claim for 1, 3, 5, 7 and 9.
 - B Assume the square leaves a remainder other than 1.
 - C Assume the claim is true for all odd integers.
 - D Write the odd integer as $2k$ for some integer k .
 - E Write the odd integer as $2k + 1$ for some integer k .
- 5 Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?
- A $n = 3k + 1, n = 3k + 2$
 - B n even, n odd
 - C n positive, n zero, n negative
 - D $n = 3k, n = 3k + 1, n = 3k + 2$
 - E n prime, n not prime
- 6 It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?
- A $f(2) < 0$
 - B $f(2) = 0$
 - C $f(2) > 0$
 - D $f(2) \leq 0$
 - E nothing about $f(2)$
- 7 Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?
- A Any three integers include one that is a multiple of 3, so the sum is a multiple of 3.
 - B Let the integers be $n - 1, n, n + 1$. Their sum is $3n$, which is divisible by 3.
 - C Let the integers be $n, 2n, 3n$. Their sum is $6n$, which is divisible by 3.
 - D The middle integer is the mean of the three, so the sum is three times an integer... but this needs the middle integer to be a multiple of 3.
 - E $1 + 2 + 3 = 6, 2 + 3 + 4 = 9$ and $3 + 4 + 5 = 12$ are all divisible by 3.

8 To prove '**if n^2 is even then n is even**' by contradiction, which assumption should be made?

- A** n is odd.
- B** n^2 is even and n is odd.
- C** n^2 is even.
- D** n^2 is odd.
- E** n is even and n^2 is odd.

9 A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$** is given with its lines scrambled.

- I** Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.
- II** Expanding, $a - 2\sqrt{ab} + b \geq 0$.
- III** Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

- A** III, II, I
- B** I, II, III
- C** II, III, I
- D** III, I, II
- E** II, I, III

10 Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

- A** If $a > b$ then $a + c > b + c$, for all real a, b, c .
- B** If $a > b$ then $a^2 > b^2$, for all real a, b .
- C** If $a > b > 0$ then $a^2 > b^2$.
- D** If $a > b$ then $a^3 > b^3$, for all real a, b .
- E** If $a > b$ then $-a < -b$, for all real a, b .

- 11 T_n denotes the n th triangular number $\frac{n(n+1)}{2}$. A student observes $T_1 + T_2 = 4$, $T_2 + T_3 = 9$, $T_3 + T_4 = 16$ and conjectures that $T_n + T_{n+1}$ is always a perfect square. Which of the following proves the conjecture?
- A $T_n + T_{n+1} = \frac{n(n+1) + (n+1)(n+2)}{2} = \frac{(n+1)(2n+2)}{2} = (n+1)^2$
- B $T_n + T_{n+1} = 2T_n + (n+1)$, which is a square.
- C Triangular numbers are always squares.
- D $T_4 + T_5 = 25$, which is a square, so the pattern continues.
- E $T_{n+1} - T_n = n+1$, so the sum is a square.
- 12 Consider the claim: **if n is a positive integer then $n^2 + n + 1$ is odd**. Which of the following is a valid proof?
- A $n^2 + n = n(n+1)$ is even, being the product of consecutive integers, so $n^2 + n + 1$ is even plus one, which is odd.
- B $n^2 + n + 1$ cannot be factorised, so it is odd.
- C For $n = 1, 2, 3$ the values are 3, 7, 13, all odd, so the claim holds.
- D Suppose $n^2 + n + 1$ is even. Then n is even, which is a contradiction.
- E If n is odd then n^2 is odd and n is odd, so $n^2 + n + 1$ is odd.